
535354 DL Team Project: Proposal Guidelines (End-to-End Staff to Jianpu Translation via Vision-Encoder-Decoder Networks)

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1 Project Overview

1.1 Profile

Track: 3. Application

Abstract and project goal: This project proposes an end-to-end deep learning approach to directly translate images of printed staff notation into numbered musical notation (Jianpu). While Optical Music Recognition (OMR) has seen significant advancements, most existing systems focus on digitizing Western notation into intermediate formats like MusicXML or MIDI. Our model employs an Encoder-Decoder architecture to bypass these intermediate steps, directly outputting an array of semantic tokens that represent Jianpu. By avoiding multi-stage pipelines and rule-based heuristics, we aim to reduce cascading errors. The overarching goal is to streamline the transcription process, creating a practical and highly efficient tool for musicians who primarily rely on numbered notation for performance and rehearsal.

TL;DR (“Too Long; Didn’t Read”): An end-to-end Encoder-Decoder OMR system that directly converts printed sheet music images into Jianpu semantic tokens, eliminating the need for intermediate format conversions.

1.2 Motivation

Converting standard five-line staff notation to numbered musical notation is a tedious yet essential task. In traditional Chinese music clubs and ensembles, Jianpu is a standard for sight-reading and performance. An automated, end-to-end tool that directly outputs readable Jianpu would eliminate hours of manual transcription, solving a major practical bottleneck and allowing musicians to focus on playing rather than score translation.

Despite its importance, this problem remains largely open. Current OMR research predominantly focuses on digitizing Western music notation. Seminal works [Calvo-Zaragoza et al., 2020] and recent open-source projects rely on multi-stage semantic segmentation coupled with heuristic rules to group notes and extract symbols. These pipelines are prone to cascading errors and are not inherently designed for cross-format translation (Staff to Jianpu). End-to-end Image-to-Sequence models for direct Jianpu generation remain a largely unexplored niche.

Developing an end-to-end model for this task presents several critical challenges:

- **Challenge 1: Bridging the 2D-to-1D Semantic Gap.** Mapping the complex 2D spatial arrangement of musical symbols (such as chords, slurs, and harmonics) on a staff to a flat 1D sequence of Jianpu semantic tokens requires robust spatial and sequence modeling capabilities.

- **Challenge 2: Direct Translation without Intermediate Representations.** Bypassing intermediate formats means the model must implicitly learn music theory and alignment rules purely from visual data, without relying on explicit heuristic programming.
- **Challenge 3: Dataset Limitations.** Since large-scale, paired datasets for Staff-to-Jianpu translation are scarce, the model must be trained effectively on synthetic or carefully curated printed sheet music, ensuring it generalizes well across different printed styles while avoiding the extreme variations of handwritten scores.

Regarding the state-of-the-art methodology, Oemer represents a strong open-source baseline for printed sheet music OMR, achieving high accuracy through UNet-based semantic segmentation [Yoyo et al., 2023]. However, for end-to-end sequence generation tasks, Transformer-based Image-to-Text architectures (such as vision-encoder-decoder models like TrOCR) represent the state-of-the-art methodology, which we aim to adapt and evaluate for this specific music translation task.

2 Problem Formulation

This project focuses on an Image-to-Sequence translation task within the domain of Optical Music Recognition (OMR). We frame the transcription of standard staff notation into numbered musical notation (Jianpu) as an end-to-end sequence generation problem. The model takes 2D images of printed sheet music as input and directly decodes them into 1D arrays of custom semantic tokens representing Jianpu characters, rhythms, and performance marks. The training process utilizes fully supervised learning, guided by paired datasets of sheet music images and their corresponding ground-truth token sequences. To address the scarcity of such paired datasets, the training signal will primarily rely on large-scale synthetic data generated by rendering digital score files.

The primary optimization objective is to minimize the sequence-level cross-entropy loss between the predicted and ground-truth token sequences, with the ultimate goal of maximizing token-level accuracy and exact sequence match rates. To maintain project feasibility, we make a few key assumptions: first, the input data is strictly restricted to printed scores (e.g., clear scans or photographs of printed sheet music), explicitly excluding the highly variable domain of handwritten scores. Second, we assume that our custom tokenization vocabulary is sufficiently expressive to capture all essential musical semantics required for music performance without structural information loss.

3 Empirical Evaluation

3.1 Performance Metrics

For our Image-to-Sequence OMR task, we plan to use the following performance metrics:

- **Symbol Error Rate (SER):** Similar to Character Error Rate (CER) in traditional OCR tasks, SER calculates the Levenshtein distance (accounting for insertions, deletions, and substitutions) between the predicted Jianpu token sequence and the ground truth. This is the most critical metric for evaluating the quality of sequence generation.
- **Sequence-level Accuracy (Exact Match Rate):** This metric calculates the percentage of completely correct musical measures or lines. It is a strict metric that ensures the entire musical context within a given window is perfectly translated without any token errors.

3.2 Baseline Methods

Since end-to-end models specifically designed for Staff-to-Jianpu translation are extremely rare, we will compare our proposed architecture with the following baselines:

- **Pipeline-based SOTA [Yoyo et al., 2023]:** Oemer, an open-source OMR system utilizing U-Net for semantic segmentation. We will evaluate its intermediate ability to extract staff symbols, which would hypothetically require an additional heuristic converter to produce Jianpu.
- **Classic Image-to-Sequence [Shi et al., 2016]:** CRNN (Convolutional Recurrent Neural Network) trained with CTC loss. This is a standard and robust baseline for visual sequence recognition tasks, which we will adapt for our custom Jianpu token vocabulary.

- **Modern Vision-Language Model [Li et al., 2023]:** TrOCR (Transformer-based Optical Character Recognition). As a pre-trained Vision-Encoder-Decoder model, it represents a strong architectural baseline for state-of-the-art end-to-end sequence generation.

3.3 Benchmark Tasks or Datasets

Because there is no standard public dataset for Staff-to-Jianpu translation, we will evaluate our method on the following custom data setups:

- **Custom Synthetic Dataset:** A large-scale training and validation set generated programmatically. We will render digital score files (e.g., MusicXML or MIDI) into both standard Staff images and their corresponding Jianpu token sequences using rendering engines (such as MuseScore or LilyPond) to provide massive paired data.
- **Real-World Music Test Set:** A manually collected and annotated test set consisting of scanned or photographed printed sheet music commonly used in music ensembles. This serves as the ultimate benchmark to evaluate our model’s generalization capability and practical utility in real-world scenarios.

4 Methodology (Optional)

Our proposed solution employs a Vision-Encoder-Decoder architecture to achieve an end-to-end translation from printed sheet music images to numbered musical notation (Jianpu) token sequences. Specifically, the model ingests a 2D image of a musical score and processes it through a visual encoder (such as a Vision Transformer or a deep Convolutional Neural Network like ResNet) to extract rich, dense spatial feature representations. These extracted visual features are subsequently passed to a Transformer-based decoder, which leverages cross-attention mechanisms to autoregressively predict a 1D sequence of custom Jianpu semantic tokens. By adopting this unified framework, the network implicitly learns both the intricate spatial relationships of standard Western musical symbols (e.g., pitches, rhythms, and chords) and the underlying music theory required for translation, entirely bypassing the need for intermediate formats or explicit, hard-coded heuristic rules.

Figure 1: description of Figure 1

5 Experimental Results

5.1 Evaluation of Baseline Methods

At the current proposal stage, we are actively constructing our custom synthetic dataset (Staff-to-Jianpu image-sequence pairs) and configuring the baseline models. We plan to conduct our preliminary experiments by evaluating the classical Image-to-Sequence baseline (CRNN) and the modern vision-language baseline (TrOCR) on our generated validation set.

To rigorously assess the baseline performance, we will report the Symbol Error Rate (SER) and the Sequence-level Exact Match Rate. The expected format for our preliminary baseline evaluation is presented in Table 1.

These preliminary results will serve as a foundation for designing and tuning our proposed end-to-end architecture. We anticipate that the Transformer-based baseline (TrOCR) will outperform the CTC-based baseline (CRNN) due to its superior sequence modeling capability, which is crucial for capturing the underlying musical syntax and alignment rules without explicit programming.

Table 1: Preliminary evaluation of baseline methods on the synthetic validation set.

Method	Symbol Error Rate (SER) ↓	Exact Match Rate ↑
Baseline 1: Oemer (with heuristic converter)	TBD	TBD
Baseline 2: CRNN (ResNet + RNN + CTC)	TBD	TBD
Baseline 3: TrOCR (ViT + Transformer)	TBD	TBD

6 Expected Contributions of Each Team Member

- BEN-LU CHEN:
- CHUN-JUI HSU:
- MENG-XI LIN:
- JIAN-AN ZHU:

References

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